

**EDO STATE COLLEGE OF AGRICULTURE AND
NATURAL RESOURCES, IGUORIAKHI, NIGERIA**

DEPARTMENT OF AGRICULTURAL TECHNOLOGY

COURSE CODE: AGT 216

COURSE TITLE: FARM SOIL MANAGEMENT

CREDIT LOAD: 3 UNITS

AGT 216: FARM SOIL MANAGEMENT

INTRODUCTION

This course provides an understanding of the general principles and practises of farm soil management. The course is introduced at both theoretical and practical level. Students are prepared to go beyond the normal in-the-classroom learning to field visits. These visits are practically used to further inculcate the classroom teachings into the students on a real-life scenario. The course aims and objectives are provided below:

Course aim and objectives

The course aims at introducing the students to the various aspects of managing the farm soil using a theoretical and practical approach to learning. On completion of this course, Students should be able to achieve the following objectives:

- A. Discuss soil conservation and have an understanding of the importance of soil conservation.
- B. Describe erosion and the various forms of erosion and its effects: This includes water and wind erosion.
- C. Understand the practices of sustainable cultivation and its principles.
- D. Appreciate the principles involved in the practice of water conservation and supply.
- E. Discuss irrigation and field drainage systems and sustainable management of irrigation and field drainage systems.
- F. Understand the effect of climate change on soil management practices.

Understanding the Importance of Soil Conservation

What is Soil Conservation?

Soil conservation refers to all the practices and techniques used to protect soil from loss, degradation, and pollution. It focuses on maintaining or improving soil fertility, structure, and productivity by preventing erosion, nutrient depletion, and other destructive processes. Soil conservation ensures that soil remains a healthy foundation for crops, livestock, biodiversity, and the environment.

Importance of Soil Conservation

Soil conservation is crucial because:

- It maintains soil fertility for sustainable agricultural productivity.
- It prevents soil erosion, keeping valuable topsoil in place.
- It supports water infiltration and reduces flooding and runoff.
- It preserves important soil organisms that enhance nutrient cycling.
- It contributes to climate change mitigation through carbon storage.
- It protects water bodies from sedimentation and pollution.
- It safeguards future food security by ensuring long-term land productivity.

Erodibility

Erodibility is the inherent susceptibility of a soil to erosion. It depends on soil properties such as texture, structure, organic matter content, and permeability. Sandy soils may be easily blown by wind, while silty soils detach and move quickly with water. Soils high in organic matter usually resist erosion better because they are well-structured and stable.

Soil Erosion

Soil erosion is the process in which the upper layer of the soil is detached and transported by agents such as water, wind, or human activities. The most fertile portion, known as topsoil, is usually lost first, reducing the soil's agricultural value and ecological functions.

Causes of Soil Erosion

Soil erosion occurs due to both natural and human-induced factors. Major causes include:

- Heavy rainfall and surface runoff
- Strong winds in dry regions
- Deforestation and removal of vegetation cover
- Overgrazing by livestock
- Poor agricultural practices such as up-slope plowing

- Construction and land disturbance activities
- Loss of organic matter leading to unstable soil structure
- Climate change increasing extreme weather events

Effects of Soil Erosion on Crop Production

Loss of topsoil leads to a long list of negative impacts on farming:

- Reduced soil fertility and nutrient loss
- Lower water-holding capacity and increased drought stress
- Poor seedling emergence due to exposed subsoil
- Increased need for fertilizers and soil amendments
- Sediment deposition in lowlands reduces usable farmland
- Overall reduction in crop yield and farm income
- Soil erosion threatens food security and sustainable agriculture if left unchecked.

Erosion Control

Erosion control refers to the use of management practices designed to reduce the detachment, movement, and deposition of soil particles. These practices aim to keep soil anchored, improve soil stability, and maintain vegetation cover.

Basic Methods of Erosion Control

Several common techniques are applied in farmlands and landscapes:

1. Agronomic Practices

- Cover cropping: planting fast-growing crops to protect soil surface
- Mulching: adding plant residues to reduce raindrop impact
- Crop rotation: improving soil structure and ground cover
- Contour farming: plowing along land contours to slow runoff

2. Mechanical/Engineering Measures

- Terracing: creating step-like structures on slopes
- Bunds and ridges: barriers that slow down runoff
- Check dams and diversion channels: directing water flow safely
- Windbreaks: trees or shrubs that reduce wind speed

3. Biological/Vegetative Measures

- Reforestation and afforestation to stabilize soils
- Grass strips or buffer zones on sloping land
- Maintaining wetlands and riverbank vegetation

Types of Soil Erosion

Soil erosion occurs in several forms depending on the agent responsible for soil displacement:

1. Water Erosion

This is the most widespread form and occurs when rainwater or flowing water detaches and transports soil particles. Key forms include:

- **Splash erosion:** Raindrops hitting bare soil, breaking down aggregates.
- **Sheet erosion:** Uniform removal of thin soil layers across large areas.
- **Rill erosion:** Small channels form due to concentrated runoff.
- **Gully erosion:** Deep channels or gullies formed when rills enlarge and become too deep to be removed by normal tillage.

2. Wind Erosion

Common in drylands and sandy soils, wind erosion occurs when wind lifts, moves, and deposits soil particles across distances.

3. Glacial Erosion

Ice sheets and glaciers drag soil and rocks along as they move, scraping the land surface.

4. Coastal Erosion

Waves and currents remove soil or sand from coastal regions, reshaping shorelines.

5. Gravity or Mass Movement

Soil moves downhill due to gravity, especially when slopes are steep or soil is saturated (landslides, soil creep).

Understanding the Nature and Effect of Wind Erosion

What is Wind Erosion?

Wind erosion is the process by which loose, dry, and finely textured soil particles are detached and carried away by strong winds. It is especially a problem in arid and semi-arid regions, open farmlands, sandy soils, and areas where vegetation cover is sparse. Wind erosion typically occurs in three stages:

1. **Saltation:** Medium-sized particles bounce or hop along the surface.

2. **Suspension:** Very fine particles are lifted high into the air and transported over long distances, contributing to dust storms.
3. **Surface creep:** Larger particles roll or slide close to the ground.

Causes of Wind Erosion

Wind erosion is encouraged by both environmental and human factors:

- Low soil moisture and prolonged drought
- Light-textured soils (sandy, loose, low organic matter)
- Sparse or completely removed vegetation cover
- Overgrazing by livestock exposing bare soil surfaces
- Tillage practices that create fine, loose soil (e.g., conventional plowing)
- Strong, persistent winds
- Land mismanagement in dryland farming systems
- Climate change increasing desertification risk

The Effect of Wind Erosion on Crop Production

Wind erosion severely impacts agriculture and food security:

- **Loss of topsoil** containing most nutrients and organic matter
- **Reduced soil fertility** leading to lower crop yields
- **Sandblasting of young crops** causing physical injury or complete plant death
- **Exposure of plant roots** making crops unstable and vulnerable to drought
- **Reduced soil water-holding capacity** making crops more drought-stressed
- **Poor seedbed quality** resulting in uneven germination
- **Sand deposition** can bury seedlings and clog irrigation systems
- **Lower farm income** due to reduced productivity and increased soil amendment costs.

In extreme cases, wind erosion contributes to land degradation, desert expansion, and forced migration, like the historic “Dust Bowl” experience.

Methods of Control and Prevention of Wind Erosion

Wind erosion control focuses on keeping the soil protected, moist, and structurally stable.

1. Vegetative Measures

- **Cover crops** like legumes or grasses to protect exposed soil
- **Windbreaks or shelterbelts** made of trees/shrubs to reduce wind speed
- **Maintaining natural vegetation** in rangelands
- **Residue mulching** to shield the soil from direct wind impact

2. Soil Management Practices

- **Conservation tillage or zero-tillage** to maintain soil aggregates
- **Increase soil organic matter** through compost, manure, or crop residues
- **Irrigation or soil wetting** in extremely dry seasons to keep soil moist and less mobile
- **Soil stabilizers** (e.g., clay, organic binders) in sandy soils to reduce mobility

3. Structural or Engineering Methods

- **Sand fences** and barriers placed across the wind direction
- **Contour ridges** to trap moving soil particles
- **Strip cropping** alternating dense crops with low cover crops to slow wind flow

Note: Wind erosion is a major threat to soil health in dry regions and open farmlands. Once valuable topsoil is lost, rebuilding fertility can take decades. Sustainable land management, vegetation protection, and improved farming practices remain the best defense. Helping farmers adopt wind erosion controls ensures stronger crop production, environmental protection, and long-term food security.

Understanding the Nature and Effect of Water Erosion

Soil supports life. Water keeps that life going. Yet water can also be one of the most powerful destroyers of soil when it is not properly managed. Understanding water erosion helps protect crops, farmland, and the environment.

What is Water Erosion?

Water erosion is the removal, detachment, and transportation of soil particles by rainfall, surface runoff, flowing rivers, or flooding. It occurs when raindrops hit exposed soil or when water flows over the land surface with enough force to move soil away from where it is needed.

Water erosion commonly occurs in stages:

- **Splash erosion:** Impact of raindrops breaks soil aggregates
- **Sheet erosion:** Thin layers of soil are removed across a wide surface
- **Rill erosion:** Small channels form as runoff concentrates in tiny streams
- **Gully erosion:** Channels become too deep to be removed by normal farming tools
- **Streambank erosion:** Flowing rivers cut into their banks and carry soil away

Once the topsoil leaves the field, rebuilding it can take hundreds of years. That is why prevention is much smarter than repair.

Causes of Water Erosion

Many natural and human activities contribute to water erosion:

I. Natural Causes

- High intensity and prolonged rainfall
- Steep land slopes that increase runoff velocity
- Soil types with loose structure or low organic matter
- Areas with low vegetation cover
- Seasonal flooding

II. Human-Induced Causes

- Deforestation exposing bare earth
- Overgrazing removing protective ground cover
- Poor land preparation such as up-slope tillage
- Construction activities and mining that disturb soil
- Improper irrigation causing runoff or soil crusting

Water erosion becomes much worse when bare soil meets heavy rainfall on sloping land. A recipe for disaster in agriculture.

Effects of Water Erosion on Crop Production

Water erosion affects both soil and crops in many damaging ways:

- Loss of fertile topsoil reduces nutrient availability
- Decline in soil organic matter leads to poorer soil structure
- Root growth becomes difficult when subsoil is exposed
- Seeds may wash away leading to poor crop establishment
- Reduced soil moisture storage causes drought stress
- Sediment carried into water bodies causes pollution
- Yield declines and farm profitability suffers
- Infrastructure such as roads and irrigation channels become buried or damaged

The long-term effect is land degradation, leaving farmland less productive each season.

Methods of Control and Prevention of Water Erosion

Effective water erosion control keeps soil covered, slows runoff, improves water infiltration, and guides water safely away from vulnerable areas.

Below are practical strategies farmers can adopt.

1. Soil Cultivation Practices

- Conservation tillage reduces soil disturbance and maintains structure
- Minimum and zero tillage increase surface residues that protect soil
- Deep tillage only when necessary and done along contour lines
- Avoid soil pulverization which creates fine particles easily eroded

Good cultivation builds a soil that resists raindrop impact.

2. Planting Direction

- Planting crops across the slope (contour farming) slows water movement
- Ridging and row orientation perpendicular to flow direction reduces runoff speed
- Strip cropping alternates erosion-resistant crops with erosion-prone ones for protection

This technique forces water to pause and infiltrate rather than rush downhill.

3. Drainage and Water Diversion

- Surface drains guide excess water away from farmlands safely
- Check dams reduce flow velocity in erosion-prone channels
- Terraces create level steps on steep slopes to trap soil and store water
- Waterways lined with grass or stones prevent channel cutting

Proper drainage prevents water from turning into a destructive force.

4. Cover Cropping and Mulching

- Cover crops protect soil from splash impact
- Mulching with crop residues reduces evaporation and strengthens structure
- Roots from cover crops hold soil together firm
- Mulch slows runoff and improves water infiltration

Soil that is covered is soil that is safe.

Note: Water erosion is a silent thief stealing soil productivity year after year. By understanding how erosion happens and using smart land management practices, farmers can protect their fields, improve crop yields, and preserve the environment for future generations. Preventing erosion is not only cheaper than restoring degraded land, it is essential for food security.

Understanding the Principles and Practices of Sustainable Cultivation

1. Concept of Sustainable Soil Management

Sustainable soil management involves using the soil in a way that maintains its productivity, biodiversity, and environmental functions for long-term agricultural use. The idea is to make sure soil resources available today remain healthy and productive for future generations.

Key principles include:

- Conserving soil structure and preventing erosion
- Maintaining soil organic matter and biological activity
- Enhancing nutrient cycling for crop use
- Avoiding contamination from chemicals and waste
- Using land according to its suitability (land evaluation)
- Ensuring continuous vegetation cover to protect soil

Sustainable cultivation prioritizes a balance between high yield, soil health, and environmental safety.

2. Causes of Soil Damage

Soil damage occurs when physical, chemical, or biological properties of soil are negatively altered. Major causes include:

Physical causes

- Excessive tillage leading to soil compaction
- Overgrazing removing protective vegetation
- Construction and land-use changes
- Erosion by wind and water

Chemical causes

- Over-application of fertilizers causing nutrient imbalance
- Soil acidity or alkalinity from prolonged chemical use
- Industrial pollution and toxic waste
- Accumulation of salts (salinization) in irrigated fields

Biological causes

- Decline in organic matter due to residue removal
- Reduction of soil microbial activity
- Continuous mono-cropping causing pest and disease buildup

These activities disturb soil structure, fertility, and ecological balance.

3. Effects of Soil Damage on Crop Growth

Damaged soil creates unfavorable conditions for seedling growth and productivity. Key effects include:

- **Reduced water infiltration** leading to waterlogging or drought stress
- **Root restriction** due to compaction, limiting nutrient uptake
- **Decline in nutrient availability** because of reduced microbial activity
- **Poor soil aeration** limiting root respiration
- **Decreased organic matter** resulting in low cation exchange capacity (CEC)
- **Increased erosion** stripping away fertile topsoil
- **Yield reduction** from nutrient deficiencies and plant stress
- **Increased pest and disease pressure** in degraded soils

Healthy soil equals healthy crops. Damaged soil struggles to support sustainable agriculture.

4. Methods of Soil Damage Repair

Improving soil health involves restoring fertility, structure, biodiversity, and water management.

Important repair strategies include:

A. Sub-soiling

Sub-soiling is deep mechanical loosening of soil below the plough layer (30–60 cm).

Benefits:

- Breaks hardpan and reduces compaction
- Improves deep root penetration
- Enhances water infiltration and drainage
- Allows better utilization of subsoil nutrients

Used mainly in fields affected by heavy machinery compaction.

B. Surface Loosening (Conservation Tillage)

This technique involves reducing the intensity of soil disturbance while maintaining crop residues on the surface.

Benefits:

- Protects soil from erosion and raindrop impact
- Reduces crusting and enhances seedling emergence
- Increases soil organic matter as residues decompose
- Supports macro- and micro-organisms that contribute to nutrient cycling

Examples: minimum tillage, ridge tillage, strip tillage.

C. Organic Matter Restoration

- Application of compost, manure, green manure, incorporation of crop residues.
- Use of biochar to improve CEC and structure.

Organic matter acts like glue, holding soil particles together.

D. Crop Rotation and Cover Cropping

- Breaks pest and disease cycles.

- Improves soil structure through diverse root systems
- Adds nitrogen when legumes are included
- Maintains ground cover to reduce erosion

E. Controlled Grazing and Vegetative Cover

- Prevents removal of too much plant cover
- Reduces compaction from livestock
- Protects topsoil from erosion

Note: Sustainable cultivation ensures long-term agricultural productivity by preventing soil degradation and actively restoring soil health. When farmers adopt soil-friendly practices such as sub-soiling, minimal tillage, crop rotation, and organic matter management, they build a strong foundation for improved crop yield, resilience, and environmental protection. Healthy soils are the backbone of food security.

PART 2: PRINCIPLES AND PRACTICES OF WATER CONSERVATION AND SUPPLY

Definition of Water Conservation

Water conservation is the efficient use and management of water resources.

Importance

- Prevent drought stress
- Improve crop yield
- Reduce water waste

Methods of Water Conservation

A. Mulching: Covering soil surface with degradable organic materials.

Benefits:

- Reduce evaporation
- Improve fertility

B. Cover Cropping

Benefits:

- Reduce erosion
- Improve soil moisture

C. Conservation Tillage

Benefits:

- Reduce water loss

SUSTAINABLE MANAGEMENT OF IRRIGATION SYSTEMS

Irrigation is the artificial supply of water to crops.

Types of Irrigation

Surface Irrigation

Examples: Flood irrigation

Sprinkler Irrigation

Modern and efficient.

Drip Irrigation

Most efficient modern system.

Benefits:

- Saves water
- Improves yield

EFFECT OF CLIMATE CHANGE ON SOIL MANAGEMENT

Effects

- Soil erosion
- Drought
- Flooding
- Nutrient loss

Solutions

- Conservation agriculture
- Agroforestry
- Organic farming

SUSTAINABLE MANAGEMENT OF FIELD DRAINAGE SYSTEMS

Definition of Drainage: Removal of excess water.

Importance

- Prevent waterlogging
- Improve root growth

Types of Drainage

- Surface drainage
- Subsurface drainage

Modern Drainage Techniques

- Drainage pipes
- Raised beds